

Circular design

Product design plays a significant role in reducing waste and emissions. Our aim is to keep products and materials in use for as long as possible, so we implement designs that allow our products to be more easily disassembled. This makes them easier to repair, refurbish, reuse and recycle.

The strategy and evolution of design at Dell

Since its introduction in FY22, [Concept Luna](#) has evolved into an inspiration and guidepost for Dell’s circular design approach. Our design teams gathered a wealth of knowledge and strategies during product prototyping that pushed the boundaries of what Dell’s sustainable technology could look like now and in the future. We continue to implement these findings in our products and further build our portfolio of innovative designs.

However, we understand that creating a positive impact on the world is not something one company can achieve alone. We engage with others – including our competitors – to drive circularity at scale and maximize impact. Our memberships in influential cross-industry partnerships like the [Circular Electronics Partnership](#) (CEP) and [NextWave Plastics](#) are examples of this collaboration in action. Hosted by the [World Business Council for Sustainable Development](#) (WBCSD), CEP brings together leaders from industry, international organizations and civil society. CEP facilitates global collaboration on an industry road map to overcome the barriers to a circular electronics industry. This includes improving product and supply chain circularity, and increasing awareness of the importance of circular electronics.

In addition to our leadership and memberships, we increased our public advocacy efforts. We participated in stakeholder education on the importance of a more circular economy in the technology sector, including a May 2023 Consumer Technology Association Circular Economy panel and a similar panel at the 2024 Consumer Electronics Show. We continue to join efforts across industries towards building a more circular economy, which aims to eliminate waste while keeping resources and materials in use longer.





Circular design principles

To accelerate the circular economy, we work towards design principles that consider sustainability throughout our process.



Sourcing sustainable materials draws on recycled, renewable or reduced carbon materials for our product designs. We work diligently to ensure the materials we source minimize our impact on the environment. Examples at Dell include low-emissions aluminum in XPS laptops, recycled aluminum with low-emissions aluminum in Latitude and Precision products and recycled copper in the cables of some power adapters.



Dematerializing and optimizing reduces needed materials or the impact of processes required to create products. Using fewer resources upfront preserves natural resources and reduces the carbon footprint of mining and manufacturing. Examples at Dell include designing for fewer internal components, lean manufacturing, passive cooling, paints, on-demand production and customer tools like virtualization.



Designing for durability creates longer-lasting products, which helps avoid or delay the environmental impact created by building new devices. Examples at Dell include improved water spill resistance, greater tolerance for shock and vibrations, wider range of temperature and humidity tolerances, better battery endurance and greater fault tolerance.



Designing for reuse and repair makes it easier to access and replace damaged or failed parts to extend product life. Reuse uses less energy than recycling, which further reduces carbon footprints. Examples at Dell include modular designs such as single-access service doors, eliminating the use of glues and creating upgradable or interchangeable parts.



Designing for parts harvesting or recycling is important to make it easy to recover parts and recycle materials from products that reach the end of their lives. Recycling materials uses less energy than processing virgin raw materials. Examples at Dell include clear labeling, easy separation of parts and material types, unified material grades and recyclable materials and working with recycling partners.



Evolving business practices by offering services and adjusting business models helps keep resources in the economy and out of landfills. Examples include as-a-Service (aaS) offerings, cascaded ownership (opportunities to remanufacture and reuse materials repeatedly), takeback programs, on-demand services and technologies to drive system change such as artificial intelligence (AI), machine learning and data analytics.